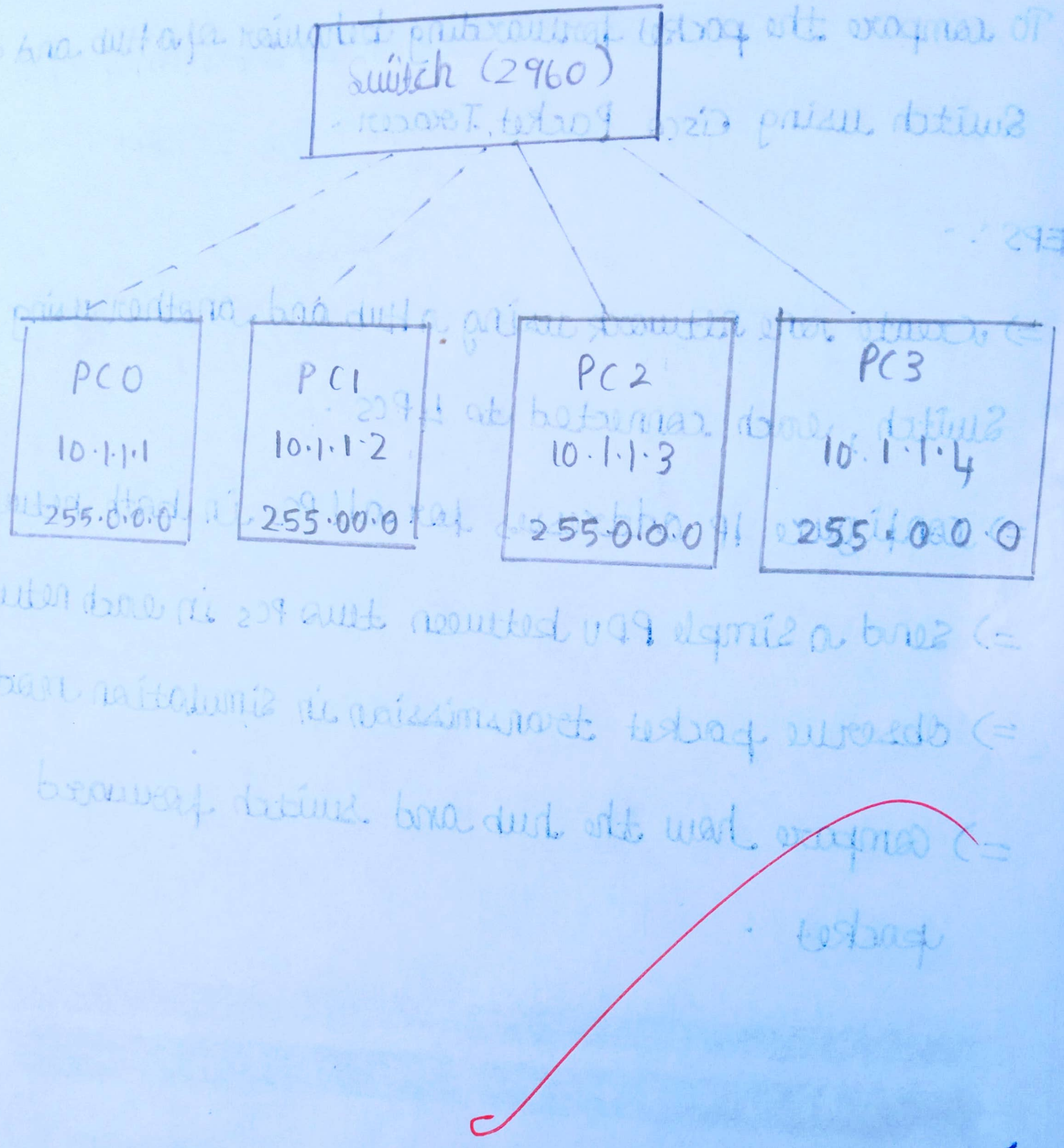

switch (2960)
PC0
10.1.1.1
255.0.0.0
PC1
10.1.1.2
255.00.0
PC2
10.1.1.3
255.0.0.0
PC3
10.1.1.4
255.0.0.0

NO:- 4
2.8.2026

LAN using a Switch and Ethernet Cables.

AIM:-

Setup and configure a LAN (Local area network) using a Switch and Ethernet cables in your lab.

What is a LAN ?

A local area network (LAN) refers to a network that connects devices within a limited area, such as an office building, school, or home. It enables users to share resources, including data, printers, and internet access. A local area network (LAN) Switch serves as the primary connecting device, managing and directing communications within the local network. Each connected device on a LAN switch can communicate directly with each other allowing for fast and secure data transfer.

Steps to Set up a LAN :-

Step 1 :-

Plan and Design an appropriate network topology taking taking into account network requirements and equipment location.

Step 2 :-

Take 4 Computers, a Switch with 8, 16, or 24 ports which is sufficient for network of these size, and 4 Ethernet Cables.

Step 3 :-

Connect the Computers to network Switch via an Ethernet cable, which is as Simple as plugging one end of the Ethernet

table.

Step4 :- Assign IP address to your PCs

1. Log on to the client computer as admin.

2. Click Network and internet connections.

3. Right click local area connection / Ethernet -> Go to Properties ->
Select Internet Protocol (TCP/IPV4) -> click on properties ->
Select use the following ip address option and assign
ipaddress.

Step5 :- configure a network switch:

1. Connect the computer to the switch: To access the
Switch's web interface, it will need to connect the
computer to the Switch.

2. Log in the web interface: Open a web browser and
enter the IP address of the switch in the address bar.
Enter the username and password to log in.

3. Configure basic setting: Once logged in, it will be
able to configure basic setting for the Switch.

4. Assign IP address as : 10.1.1.5, Subnet mask 255.0.0.0

Step 6 :- Check the connectivity between switch and other
machine by using ping command in the command prompt
of the device.

Share it with everyone on the same LAN.

Step 8 :- Try to access the shared folder from others computers of the network.

RESULT :

A LAN was successfully configured using a switch and ethernet cables. IP addresses were assigned correctly. Communication was verified using the ping command, and file sharing was successfully demonstrated. Computer CAN.